LESSON 8.3

# Write and Solve Equations Using Multiplication or Division

**Start here** · Words, worked example, and sentence starters

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can solve one-step multiplication and division equations using inverse operations.

**WHY THIS MATTERS** Detective Park discovered that 3 identical locked boxes contain a total of 21 pieces of evidence. She writes the equation 3x = 21 to find how many pieces are in each box. How can she solve for x?

START HERE

## Words for this lesson

Read each word before you begin. Say it out loud.

| WORD | WHAT IT MEANS | EXAMPLE |
|---|---|---|
| **Inverse operation** *Spanish: Operación inversa* | Two math actions that undo each other, like × and ÷. | × and ÷ |
| **Multiply** *Spanish: Multiplicar* | To add equal groups again and again. | 3 × 7 = 21 means 3 groups of 7 equals 21 |
| **Divide** *Spanish: Dividir* | To split into equal groups. | 20 ÷ 4 = 5 |
| **Isolate** *Spanish: Despejar* | To get the letter by itself on one side. | get p by itself |
| **Coefficient** *Spanish: Coeficiente* | The number in front of a letter, like the 3 in 3x. | the 3 in 3x |
| **Solution** *Spanish: Solución* | A number that makes the equation or inequality true. | x = 3 is the solution of 6x = 18 because 6 × 3 = 18. |

HOW IT WORKS

## Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

**5 boxes each hold the same number of pencils, 45 in all. Solve 5b = 45.**

1. Divide both sides by 5: 5b ÷ 5 = 45 ÷ 5.
2. Simplify: b = 9.
3. Check: 5 × 9 = 45.

**Answer:** Each box holds 9 pencils.

NOW YOU TRY

## Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

**Solve x ÷ 4 = 9. Is x multiplied or divided?**

1. _____
2. _____
3. _____

**Answer:** _____

SAY IT AND WRITE IT

## Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The variable is multiplied by _____ , so I _____ both sides by it.
- My equation is _____ because each group holds _____ .
- The solution _____ means _____ in this problem.
- I checked my answer by multiplying _____ and _____ .

**Word bank** equation | coefficient | multiply | divide | inverse operation | isolate | both sides | equal groups | solution | check | undo | balance

WATCH OUT!

## A common mistake

A common mistake in Solve Multiplication and Division Equations is applying the SAME operation again instead of the inverse

LESSON 8.3

# Write and Solve Equations Using Multiplication or Division

**Version A** · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can solve one-step multiplication and division equations using inverse operations.

**WHY THIS MATTERS** Detective Park discovered that 3 identical locked boxes contain a total of 21 pieces of evidence. She writes the equation 3x = 21 to find how many pieces are in each box. How can she solve for x?

## PART 1 Understand the idea

models and meaning

**1** **Write the name of the correct group on each line.**

Sort each equation by the inverse operation needed.

**Groups:** Divide both sides | Multiply both sides

_____ 4x = 32

_____ 7n = 56

_____ 10p = 80

_____ x / 6 = 3

_____ m / 2 = 15

_____ y / 9 = 4

**HINT** If the variable is being multiplied (like 4x), divide to undo it.

## PART 2 Apply it

real-world problems

**2** **Circle the letter of the best answer. Show how you know.**

Solve: 6x = 42

(A) x = 7

(B) x = 36

(C) x = 48

(D) x = 252

**HINT** 6x means 6 TIMES x — the variable is being multiplied by 6.

SHOW YOUR WORK

**3** **Circle the letter of the best answer. Show how you know.**

Solve: n / 3 = 8

(A) n = 3
(B) n = 5
(C) n = 11
(D) n = 24

**HINT** n is being DIVIDED by 3. Which operation undoes division?

**SHOW YOUR WORK**

**4** **Circle the letter of the best answer. Show how you know.**

Solve: 9m = 63

(A) m = 7
(B) m = 54
(C) m = 72
(D) m = 567

**HINT** 9m means 9 times m — that's multiplication, even with no × symbol written.

**SHOW YOUR WORK**

**5** **Circle the letter of the best answer. Show how you know.**

Solve: y / 7 = 5

(A) y = 2
(B) y = 7
(C) y = 12
(D) y = 35

**HINT** y is split into 7 equal parts. To rebuild y, you need the opposite of dividing.

**SHOW YOUR WORK**

## PART 3 Explain and correct

reasoning

**6** **Find the mistake. Explain it, then show the correct work.**

### Spot the Common Mistake

1. **Equation** 6x = 42
2. **Choose operation** To undo it, I will multiply both sides by 6.
3. **Apply to both sides** 6x × 6 = 42 × 6
4. **Simplify** x = 252

**Which step has the mistake? Explain what went wrong, then show the correct work.**

**HINT** The student multiplied again instead of dividing. Does multiplying by 6 get x alone?

CORRECT WORK

**Correct answer:** _____

EXPLAIN YOUR THINKING

### How do you decide whether to multiply or divide to solve an equation?

**SENTENCE STARTER** When I see ___ in the equation, I know to ___ both sides because the inverse of ___ is ___.

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 8.3

# Write and Solve Equations Using Multiplication or Division

**Version B** · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can solve one-step multiplication and division equations using inverse operations.

**WHY THIS MATTERS** Detective Park discovered that 3 identical locked boxes contain a total of 21 pieces of evidence. She writes the equation 3x = 21 to find how many pieces are in each box. How can she solve for x?

## PART 1 Understand the idea

models and meaning

**1** **Complete the table. Show how you found each value.**

Solve each equation, show your work, and check.

| Equation | Work | Solution | Check |
|---|---|---|---|
| 8m = 56 | | | |
| y / 6 = 11 | | | |
| 12k = 96 | | | |

SHOW YOUR WORK

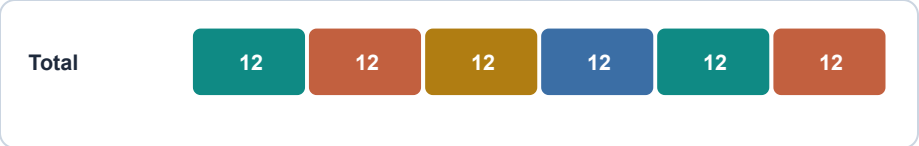
**2** **Decide whether each pair balances. Check Yes or No, then explain.**

Solve 5x = 40 using the balance scale.

| Left side | Right side | Balanced? |
|---|---|---|

EXPLAIN YOUR CHOICE

**3** Use the model to solve. Show your work.




A detective found 72 pieces of evidence and divided them equally into 6 bags. Use the bar model to show how many are in each bag.

SHOW YOUR WORK

**Answer:** _____

## PART 2 Apply it

real-world problems

**4** Circle the letter of the best answer. Show how you know.

A detective splits 54 clues equally among 9 case files. The equation 9c = 54 represents this. What is c?

(A) c = 6
(B) c = 9
(C) c = 45
(D) c = 63

SHOW YOUR WORK

## PART 3 Explain and correct

reasoning

**5** Find the mistake. Explain it, then show the correct work.

### Spot the Common Mistake

1. **Equation** n / 4 = 12
2. **Choose operation** n is in a fraction, so I will divide both sides by 4.
3. **Apply to both sides** n / 4 ÷ 4 = 12 ÷ 4
4. **Simplify** n = 3

**Which step has the mistake? Explain what went wrong, then show the correct work.**

CORRECT WORK

**Correct answer:** _____

EXPLAIN YOUR THINKING

**How do you decide whether to multiply or divide to solve an equation?**

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 8.3

# Write and Solve Equations Using Multiplication or Division

**Challenge** · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can solve one-step multiplication and division equations using inverse operations.

**WHY THIS MATTERS** Detective Park discovered that 3 identical locked boxes contain a total of 21 pieces of evidence. She writes the equation 3x = 21 to find how many pieces are in each box. How can she solve for x?

## PART 1 Understand the idea

models and meaning

**1** **Complete the table. Show how you found each value.**

Solve each equation, then write and solve a related equation that uses the inverse operation.

| Equation | Solution | Related Equation | Related Solution |
|---|---|---|---|
| 4x = 36 | | | |
| x / 7 = 8 | | | |

SHOW YOUR WORK

## PART 2 Explain and correct

reasoning

**2** Find the mistake. Explain it, then show the correct work.

### Find the Equation Error

1. **Equation** x / 5 = 12
2. **Identify inverse** Inverse of division is division
3. **Apply to both sides** x / 5 / 5 = 12 / 5
4. **Simplify** x = 2.4

**Which step has the mistake? Explain what went wrong, then show the correct work.**

CORRECT WORK

**Correct answer:** _____

**3** Answer in complete sentences.

A bakery packed muffins into boxes of 8. They made 96 muffins total. Write an equation, solve it, and explain why your answer is reasonable. Then write a different situation that uses division and has the same answer.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

**How do you decide whether to multiply or divide to solve an equation?**

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help